Chat with Logistics Document (RAG)

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1.	REET SOOD	EN22CS301796
2.	PARTH AKHAND	EN22CS301687
3.	ROSHNI RAWAT	EN22CS301826
4.	PRAKRATI RASANIYA	EN22CS301722
5.	RIYANSHI JAIN	EN22CS301816
6.	RISHI YADAV	EN22CS301806

Group Name: Group 02D7

Project Number: GAI-25

Industry Mentor Name: Aashruti Shah Ma'am

University Mentor Name: Dr. Hemlata Patel Ma'am

Academic Year: 2022-26

Table of Content

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement & Objectives

1. Problem Statement

In the logistics and supply chain industry, organizations handle a vast amount of unstructured data in the form of documents such as shipment reports, operational manuals, invoices, and bills of lading. These documents often contain critical information, but extracting specific insights from them is a tedious and time-consuming process.

Traditional search systems rely on keyword-based matching, which lacks contextual understanding. As a result, users often fail to retrieve the most relevant information. On the other hand, modern AI-based systems like Large Language Models (LLMs) can generate human-like responses but may produce incorrect or misleading information when they are not grounded in actual data.

This creates a major challenge: how to build a system that can both understand document context and provide accurate, reliable answers without hallucination.

To address this problem, this project proposes a Retrieval-Augmented Generation (RAG) system that combines the strengths of semantic retrieval and language models to deliver precise and context-aware responses based strictly on the uploaded logistics documents.

2. Project Objectives

The primary objective of this project is to design and implement an intelligent document interaction system that allows users to extract meaningful information from logistics documents efficiently.

The specific objectives include:

- To develop a system capable of processing unstructured PDF documents
- To implement text chunking techniques for efficient handling of large documents
- To convert textual data into vector embeddings for semantic understanding
- To utilize a vector database for efficient storage and retrieval of embeddings
- To implement similarity-based retrieval for fetching relevant document content
- To integrate a Large Language Model (LLM) for generating responses

- To ensure that responses are grounded in the provided documents to avoid hallucination
- To build an interactive and user-friendly interface for seamless user experience

3. Scope of the Project

The scope of this project is focused on developing a domain-specific document intelligence system for logistics-related data.

The system supports:

- Uploading and processing of PDF documents
- Semantic search using embeddings
- Context-based question answering
- Interactive chat-based interface

However, the project has certain limitations:

- It currently supports only PDF document format
- It is designed to run locally and is not deployed on cloud infrastructure
- It is optimized for small to medium datasets and may require enhancements for large-scale enterprise use

Despite these limitations, the system serves as a strong foundation for building scalable and enterprise-ready AI-powered document solutions.

Proposed Solution

1. Key Features

The proposed system offers several key features that make it efficient and user-friendly.

It allows users to upload logistics-related PDF documents and process them automatically. The system intelligently breaks down large documents into smaller segments to ensure efficient processing. These segments are then converted into embeddings, enabling semantic understanding of the content.

A vector database is used to store these embeddings, allowing fast and accurate retrieval of relevant information. When a user submits a query, the system retrieves the most relevant content and generates a response using a Large Language Model.

Additionally, the system includes a chat-based interface, making it easy for users to interact with the documents in a conversational manner.

2. Overall Architecture / Workflow

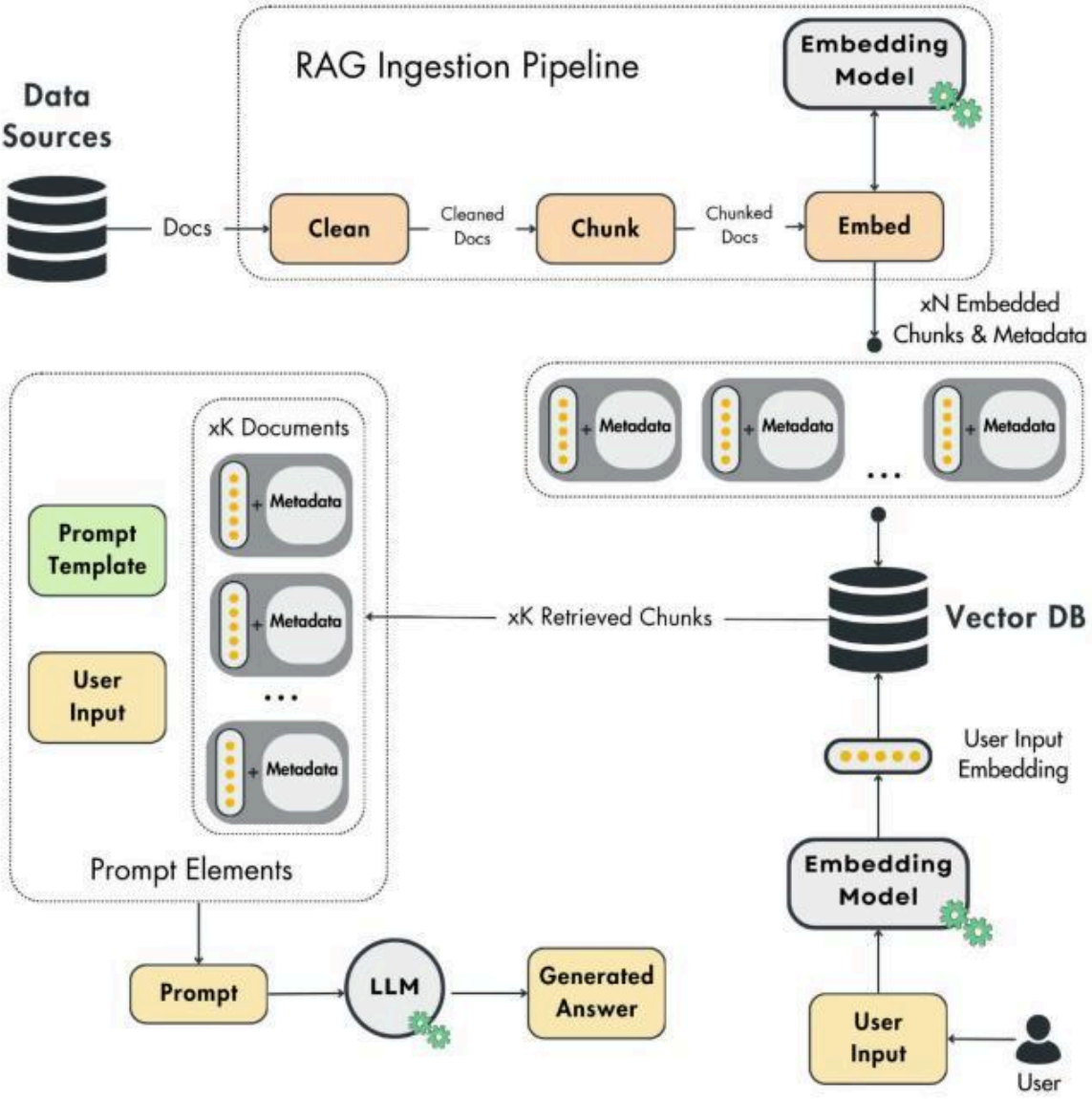
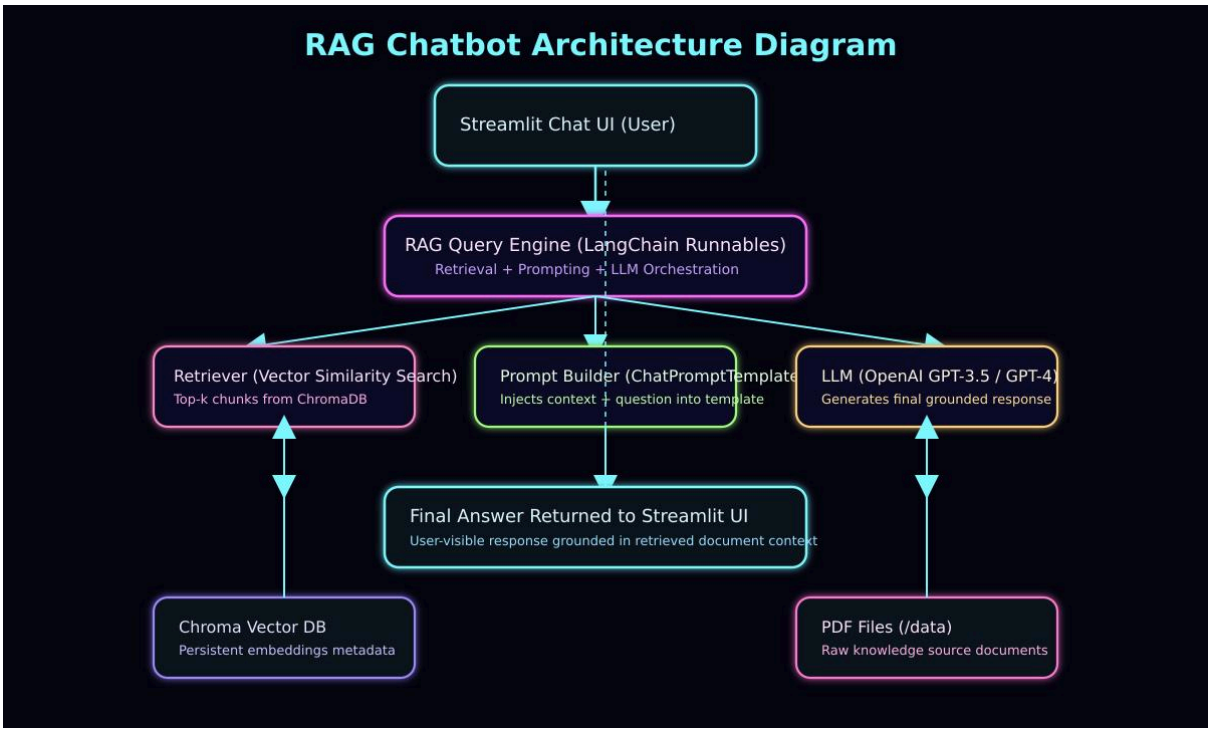


Figure 4.1: Vanilla RAG architecture



The system follows a structured pipeline that ensures efficient processing and accurate results.

Initially, the user uploads logistics documents through the user interface. These documents are stored locally and processed using a document loader. The extracted text is then divided into smaller chunks using a recursive chunking method. This step ensures that large documents can be handled effectively without losing context.

Each chunk is then converted into a numerical representation known as an embedding. These embeddings capture the semantic meaning of the text and are stored in a vector database. This database acts as a repository for all processed document information.

When a user asks a question, the system converts the query into an embedding and performs a similarity search against the stored embeddings. The most relevant chunks are retrieved and combined with the user's query to form a structured prompt.

This prompt is then passed to a Large Language Model, which generates a response based strictly on the retrieved context. The final response is displayed to the user through the interface.

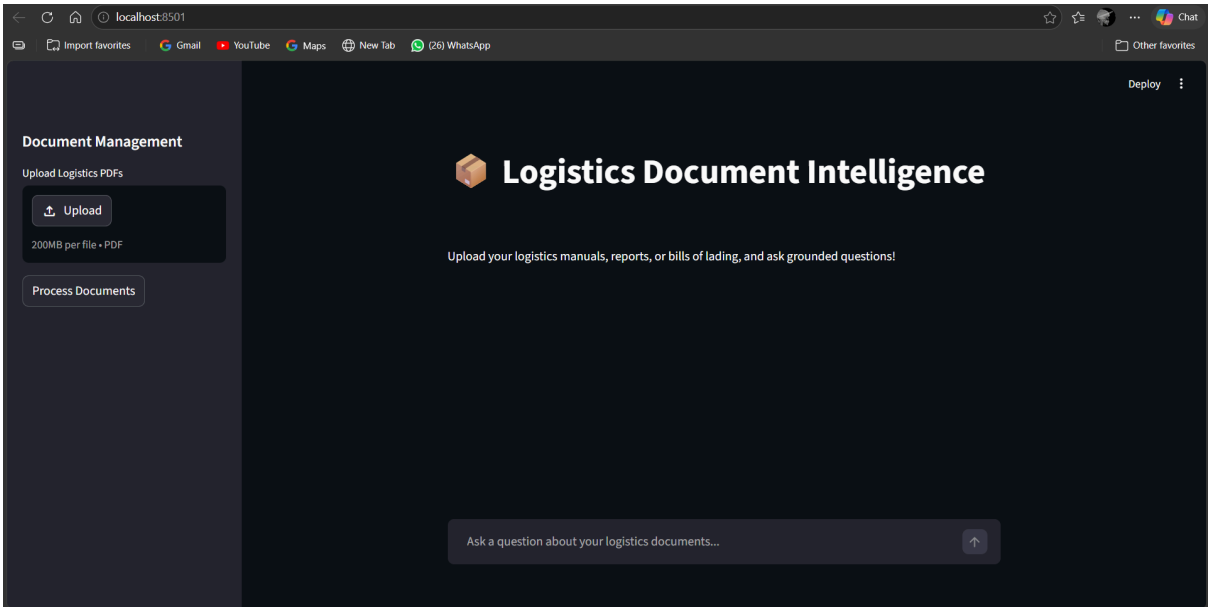
3. Tools & Technologies Used

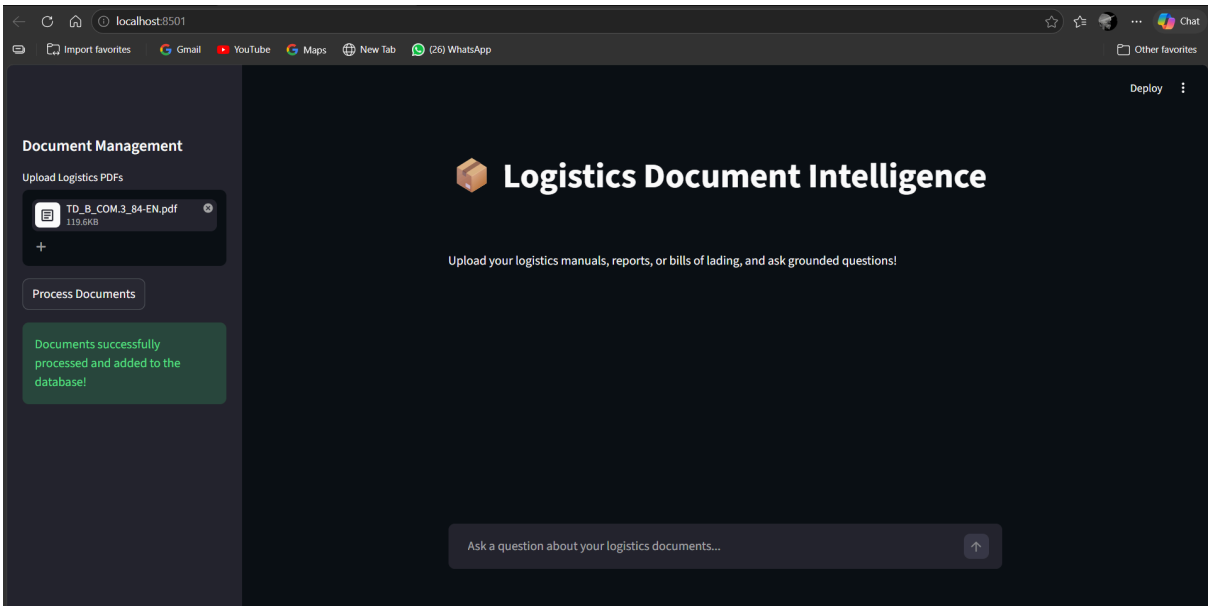
The system is built using a combination of modern tools and technologies:

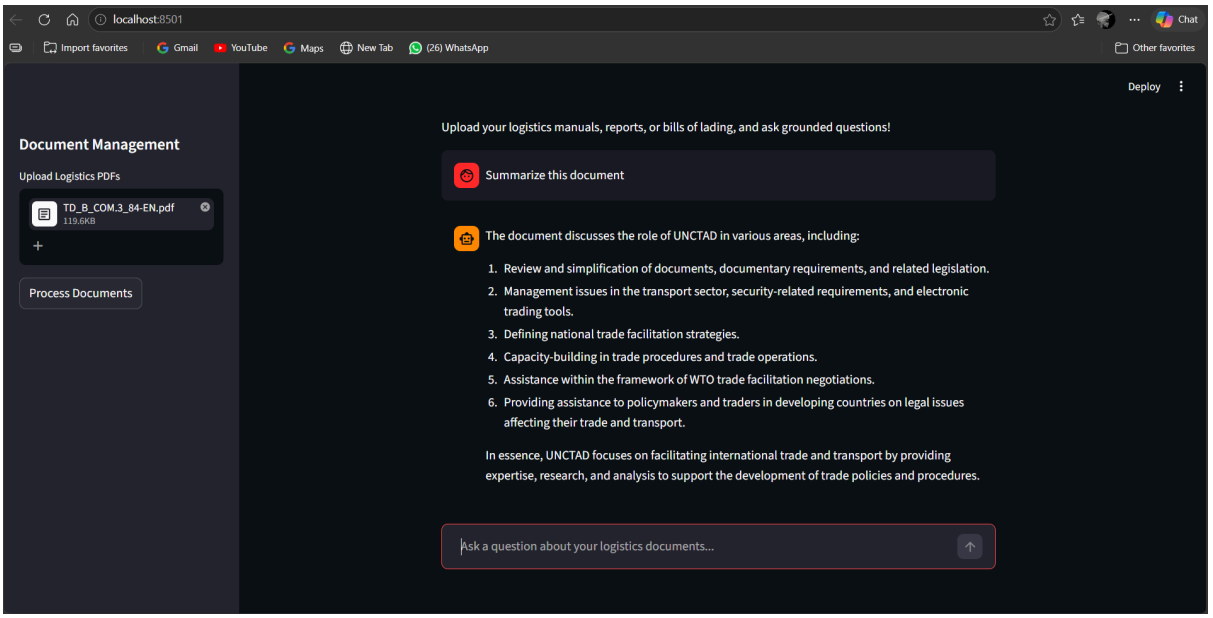
- **Python** is used as the primary programming language due to its extensive support for AI and machine learning libraries.
- **Streamlit** is used to develop the frontend interface, providing an interactive and user-friendly experience.
- **LangChain** is used to manage the document processing pipeline, including loading, chunking, and prompt handling.
- **HuggingFace Embeddings (all-MiniLM-L6-v2)** are used to convert text into vector representations.
- **ChromaDB** is used as the vector database to store embeddings and perform similarity search.
- **Groq API** is used to access the LLaMA 3.1 language model for generating responses.

Results & Output

1. Screenshots / Outputs







The system provides a simple and intuitive interface where users can upload documents and interact with them through a chat window.

Example outputs include:

- Successful document upload and processing
- User queries and corresponding responses
- Context-based answers generated by the system

2. Reports / Dashboards / Models

The system generates and maintains several internal components:

- A vector database containing embedded document chunks
- Processed document data stored locally
- A structured pipeline for retrieval and generation

While the system does not include visual dashboards, it provides real-time responses through the chat interface, which acts as the primary output mechanism.

3. Key Outcomes

The project successfully achieves its objectives by:

- Implementing a functional RAG-based system
- Enabling semantic understanding of logistics documents
- Providing accurate and context-aware responses
- Reducing hallucination through strict prompt design
- Demonstrating the practical use of AI in document intelligence

Conclusion

The Chat with Logistics Document (RAG) system demonstrates how modern AI techniques can be effectively applied to solve real-world problems in the logistics domain. By combining document processing, semantic search, and language models, the system provides a powerful tool for extracting meaningful insights from unstructured data.

The use of embeddings and vector databases allows efficient retrieval of relevant information, while the integration of a Large Language Model ensures that responses are coherent and contextually accurate. The system also addresses the common issue of hallucination by enforcing strict grounding in document content.

Overall, the project highlights the potential of Retrieval-Augmented Generation systems in improving efficiency, accuracy, and user experience in document-based applications.

Future Scope & Enhancements

The project can be further enhanced in several ways to improve its functionality and scalability.

Future improvements may include:

- Supporting additional document formats such as DOCX and TXT
- Deploying the system on cloud platforms for wider accessibility
- Integrating advanced LLMs for improved reasoning capabilities
- Adding authentication and multi-user support
- Enhancing the user interface for better usability
- Implementing real-time document updates
- Incorporating analytics and dashboard features

These enhancements can transform the system into a fully scalable enterprise solution.